REQUEST FOR PROPOSAL

For

SUPPLY, INSTALLATION, CONFIGURATION, TESTING AND COMMISSIONING

Of

Scalable Software Defined Distributed Native Object and File Systems Scale-Out Storage Solution



Satish Dhawan Space Centre SHAR

Indian Space Research Organization

Sriharikota – 524 124, A.P

RFP for Scalable Software Defined Distributed Native Object and File Systems Scale-Out Storage Solution:

Supply, Installation Configuration Testing and Commissioning

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I. INTRODUCTION

Satish Dhawan Space Centre SHAR at Sriharikota is one of the best-known names among the spaceports of the World today. India's space vehicles fly from here giving an assured access to space for domestic and as well as, commercial satellites. Diverse kinds of space missions for launching remote sensing, communications and scientific satellites are accomplished from here. The space centre, SHAR (Sriharikota Range), was renamed as 'Satish Dhawan Space Centre SHAR' on September 5, 2002, in memory of Prof. Satish Dhawan, former chairman of the Indian Space Research Organisation (ISRO).

Sriharikota is a coastal island located in the backwaters of Pulicat lake, around 80km north of Chennai.

II. ARRANGEMENT OF BID:

Proposals are invited from the interested bidders for the following in two-part bid.

Scope of Works

Supply, Installation, Configuration, Testing and Commissioning of Scalable Software Defined Distributed Native Object and File Systems Scale-Out Storage Solution with the specified configuration as per the detailed specifications in section VII.

The quote for the above requirements should be quoted in two-part bid.

Part-1 Technical and Commercial bid

Part-2 Price bid.

Part-I – Techno-commercial Bid

This part shall contain only the technical details and specifications together with technical catalogues. All commercial conditions shall also be indicated in this part. Deviations, if any, to our specifications shall be brought out very clearly. Tenderers shall mention point-wise confirmation with regard to Technical Specifications given in our Enquiry. No price details shall be mentioned in this bid and shall not upload the details of price along with the techno-commercial bid.

Part-II – Price Bid

In view of Two-Part Tender, the Offers submitted contrary to above instructions will be summarily rejected.

1) In case, the tenderer is not interested to participate in the tender, the tenderer shall submit regret letter giving reasons, failing which future enquiries will not be sent.

2) **Offer Validity**: The validity of the offers / tenders should be 120 days from the date of opening of the tenders. **Tenders with offer validity less than the period mentioned above, will not be considered for evaluation.**

3) **GST** (GSTN No. of SDSC SHAR: 37AAAGS1366J121) - With effect from 01.07.2017, Goods & Service tax (GST) has been implemented. Accordingly, the suppliers have to indicate clearly the % (full rate of tax) of GST applicable against each item in their offers.

4) **Customs Duty** – As per notification No.05/2018 of Customs department dated 25th January, 2018, ISRO is eligible only for partial exemption of Customs Duty. The reduced rate of CD is 5% plus cess@10% on CD thereon and IGST @5%. Suppliers are requested to submit their bid by loading these elements wherever, it is applicable. In case of an order, we will issue CD exemption certificate to avail the exemption. The actual amount will be reimbursed against submission of documentary evidence.

   In case of tenderers offering items considering customs duty exemption, they should also indicate the bill of materials and price, separately, with Customs Duty component and terms and conditions thereto.

5) **Liquidated Damages** - In all cases, delivery schedule indicated in the Purchase Order/Contract is the essence of the contract and if the party fails to complete the scope of work within the delivery schedule, Liquidated Damages will be levied @ 0.5% per week or part thereof the undelivered portion of the contract, subject to a maximum of 10% of total order value.

6) **Performance Bank Guarantee** - Performance Bank Guarantee for 3% of the order value should be furnished in the form of Bank Guarantee from nationalized/scheduled bank or by Demand Draft valid till warranty period plus sixty days.

7) **Security Deposit** – Security Deposit for 3% of the order value is mandatory. Party shall furnish the Security Deposit in the form of Bank Guarantee from nationalized/scheduled/First class international reputed bank or by Demand Draft valid till completion of the total scope of work plus sixty days for faithful execution of the contract.

8) The Department will have the option to consider more than one source of supply and final orders will be given accordingly.

9) The bidders should note that conditional discounts would not have any edge in the evaluation process of tenders.

10) Non-acceptance of any conditions wherever called for related to Guarantee/ Warranty, Performance Bank Guarantee, Security Deposit, Liquidated damages, then such bids are liable for disqualification.

11) Wherever installation/ commissioning is involved, the guarantee/warranty period shall reckon only from the date of installation and commissioning.

12) **Purchase/Price Preference will be extended to the MSMEs under the Public Procurement Policy for MSMEs formulated under the Micro, Small and Medium Enterprises Development Act, 2012 and instructions issued by Government of India from time to time. Vendors who would like to avail the benefit of MSME should clearly mention the same and submit all the documentary evidences to substantiate their claim along with tender itself.**

13) The drawings, specifications, end use etc., given by the Centre/Unit along with the tender enquiry are confidential and shall not be disclosed to any third party.

14) **SPECIAL CONDITIONS FOR SUBMITTING QUOTATIONS IN FOREIGN CURRENCY BY THE INDIAN AGENTS**

The Tenderer should submit the following documents/information while quoting: -

a) Foreign Principal's proforma invoice/quote indicating the commission payable to the Indian Agent and nature of after sales service to be rendered by the Indian Agent.

b) Copy of Agency agreement with the Foreign Principal and the Indian Agent, precise relationship between them and their mutual interest in the business.

c) Registration and item empanelment of the Indian Agent.

d) Agency Commission will be paid only in Indian Currency.

e) Compliance of the tax laws by the Indian Agent.

15) **High Sea Sales** - Against High Sea Sale transactions:

a. Offers shall be on all-inclusive basis including delivery up to Sriharikota at the risk and cost of the supplier. Customs Clearance is the responsibility of the supplier and at his cost and risk.

b. 100% payment will be made within 30 days after receipt and acceptance of the items at our site.

c. Customs Duty Exemption Certificate and other relevant documents required for Customs clearance will be provided.

d. High Sea Sales Agreement furnished by the supplier in accordance with the terms and conditions of our purchase order will be signed and issued by SDSC SHAR.

e. The agent and principal shall not quote for the same tender.

16) The following information/ documents are to be submitted wherever applicable.

1. Product Literature
2. Core banking account number, RTGS Details
3. PAN No. in quotation and invoices

4. GST Registration details.
5. In case of MSME, registration details / documents from Competent Authority.

17) EXCLUSION OF TENDERS

The following tenders shall be summarily rejected from the procurement process: -

(a) Tenders received from vendors who have not qualified in terms of their registration.

(b) Tenders received against publishing of a limited tender in the CPP portal.

(c) Tenders of vendors who have been removed from the vendor list or banned/debarred from having business dealings.

(d) Unsolicited tenders from vendors.

(e) The tenders which materially depart from the requirements specified in the tender document or which contain false information.

(f) The tenders which are not accompanied by the prescribed Earnest Money Deposit.

(g) The tenders of vendors who have not agreed to furnish Security Deposit or Performance Bank Guarantee.

(h) The validity of the tenders is shorter than the period specified in the tender enquiry.

(i) The tenders received from vendors or their agents or anyone acting on their behalf, who have promised or given to any official of the Centre/Unit/Department, a gratification in any form, or anything of value, so as to unduly influence the procurement process.

(j) The tenders received from vendors, who, in the opinion of the Centre/Unit, have a conflict of interest materially affecting fair competition.

(k) The tenders received from Indian agents on behalf of their foreign Principals/OEMs (in cases where the Principals/OEMs also submit their tenders simultaneously for the same item/product in the same tender).

(l) In case two or more tenders are received from an Indian agent on behalf of more than one foreign Principal/OEM, in the same tender for the same item/product.

(m) If a firm quotes 'NIL' charges / consideration, the bid shall be treated as un-responsive and will not be considered.

18) DELIVERY SCHEDULE

S.No	Mile Stone	Expected date of completion
1	PO release	T_0
2	LC Opening T_1	T_0 + 4 weeks
3	Supply of all items T_2	T_1 + 8 weeks
4	Site acceptance test T_3	T_2 + 4 weeks

5	Configuration & Installation to SDSC SHAR requirements	T_3 + 8 weeks

19) DELIVERY TERM In case of order in Indian Rupees: FOR-SDSC SRIHARIKOTA/SULLURUPETA. In case of Foreign orders, FOB/ FCA

20) MODE OF PAYMENT: All the payments due to Bidder shall be made through RTGS/NEFT. Bidder shall submit the banker details and payments can also be made through Electronic Clearance System (ECS).

21) TERMS OF PAYMENTS:

1) Our standard payment term will be 100% within in 30 days against the Supply, Installation, configuration, testing and implementation of system.

If Bidder prefers the following payment may be considered:

2) Our payment term is 80% of supply cost including taxes and transportation cost within 30 days after receipt and acceptance of sub-systems at SDSC SHAR. 20% of supply portion and 100 % of erection and commission charges along with taxes will be paid after successful commissioning and handing over of the material at SDSC SHAR on submission of Performance bank guarantee.

3) Wherever advance payment is requested, Bank Guarantee (BG) from any nationalized bank/ Scheduled bank should be furnished. The validity of BG should be till completion of scope of work under the contract. In case of advance payments, if the bidder is not supplying the material within the delivery schedule, interest will be levied as per the bank lending rate plus 2% penal interest.

4) Interest will be loaded for advance payments/stage payments as per the lending rate of bank and will be added to the landed cost for comparison purpose. In case of different milestone payments submitted by the parties, a standard and transparent methodology like NPV will be adopted for evaluating the offers.

5) In case of foreign orders, Payment term shall be Sight Draft / 90% by irrevocable letter of credit and balance 10% by Wire transfer after receipt and acceptance of the ordered item.

6) Bidder shall indicate the Foreign exchange component considered in their offer. The foreign exchange component shall be included in the lump sum price quoted / unit rates quoted. The foreign exchange component shall be limited to the amount indicated.

7) Foreign exchange rate prevailing on the date of import/payment shall be considered for adjustment of foreign exchange outgo / lump sum price over the base price and foreign exchange rate considered in the price bid / as indicated in the contract / purchase order against documentary evidence indicating the prevailing conversion rates for foreign currencies.

8) The base rate shall be considered as on the date of opening the price bids.

9) Payment for onsite support services: Payment will be made on quarterly basis after successful completion of work assigned and certified by Engineer-in-charge and approved by next higher authority. One year period will start from the date of commissioning of the system. Bidder has to provide one onsite engineer.

22) <u>PACKING AND FORWARDING</u>:

The bidder shall arrange to have all the material suitably packed as per the standards & statutes and as specified in the contract. Unless otherwise provided for in the contract, all containers (including packing cases, boxes, tins, drums, and wrappings) used by the bidder shall be non-returnable. All packing and transport charges, transit handling costs, transit risk coverage and transport fees of agents employed at the place of delivery or elsewhere, shall be deemed included in the price to be paid to the bidder.

23) <u>WARRANTY</u>:

The bidder shall provide 36 months' warranty for the entire system for any defect liability from the date of installation, after final official handing over at the bidders cost. During this period bidder has to provide and adhere to the following:

1. This period shall include maintenance, replacement of defective/failed parts at free of cost.
2. Bidder has to attend unlimited breakdown calls.
3. Department will not provide any transport/accommodation for this purpose.
4. Upon oral or written notification of defect in or malfunctioning of the goods during the warranty period which require corrective action, bidder shall send the necessary personnel to job site to supervise and assume responsibility for repair and/or replacement, if necessary, of the defective goods or material at his own cost. If bidder doesn't respond, within seven days after receipt of notification, the department has got every right to resolve reported problem and department may do so at the cost and expense of the bidder. Bidder shall reimburse to the department all expenses incurred by the department to repair or replace malfunction or non-conforming goods and forfeit the performance security.
5. Department has no obligation to discover defects, patent or otherwise, and this shall be sole responsibility of bidder. Inspection and clearance for shipment by department's inspectors or inspection agency appointed by the department shall not relieve bidder of any of bidders obligation and duties under the terms and conditions herein.
6. Where defects in items are remedied under warranty, the period for which the warranty operates shall be extended by such period, as the items were not available to the department. Where defect items are replaced by the new ones, the full warranty period stipulated in the contract shall apply to such replacement items as from the date of their delivery.
7. Bidder shall obtain similar warranty from each of his sub-bidders. However, the overall responsibility shall lie with the bidder.

24) <u>ARBITRATION</u>:

In the event of any question, dispute or difference arising under these conditions of any condition in the purchase order or in connection with this contract, (except as to any matters the decision of which is specially provided for by these conditions) the same shall be referred to the sole arbitration of the Director,SDSC SHAR. The arbitration shall

be conducted with in India as per Arbitration & Conciliation Act 1996 and the applicable language shall be English. The arbitrator may be a Retired Judge or a Government Servant or any other responsible person, that he had to deal with matter to which the contract is related or that in the course of his duties, he has expressed views on all or any of the matters in dispute or differences. The award of the Arbitrator shall be final and binding on the parties of this contract.

25) FORCE MAJEURE:

Should a part or whole work covered under this agreement be delayed due to reasons of Force Majeure which shall include legal lockouts, strikes, riots, civil commotion, fire accident, quarantines, epidemic, acts of God and Government, fright embargoes, the completion period for work, equipment referred to in this agreement be extended by a period not in excess of the duration of such Force Majeure. The occurrence shall be notified by either party within reasonable time.

26) APPLICABLE LAW AND JURISDICITION

The laws of India shall govern this contract for the time being in force. The Courts of Andhra Pradesh only shall have jurisdiction to be with and decide any legal matters or disputes what so ever arising out of the contract.

General Conditions:

	General
1	Performance Bank Guarantee (PBG) at 3% of the quoted value valid for 36 months plus two months should be provided.
2	SDSC SHAR will have option to extend the warranty period if the performance is not satisfactory during warranty period.
3	Supplier should use his own means of transport for internal movements inside SDSC SHAR.
4	If the quotation is from anybody other than Manufacturer, a letter from the manufacturer authorizing the agency to quote and provide services related to supply, installation and warranty are to be provided.
5	The selected party should follow the security procedures applicable at SDSC SHAR.
6	The Price basis for all the items should be for supply, installation, configuration and warranty support at SDSC SHAR.
7	As per Notification No. 05/2018 CUSTOMS Dt. 25.01.2018 ISRO is eligible to pay reduced rate of customs duty of 5% + cess@10% on CD thereon and IGST @5%. The offer should clearly indicate whether it is local supply or direct import supply or High Sea Sales supply. The offer should be clear about any other taxes and duties.
8	SDSC SHAR has right to place order on a single party for all the items for ease of supply and service (overall lowest will be considered).

9	The quotation should accompany the technical brochure. If not provided the quotation may be rejected.

III. MINIMUM CRITERIA FOR QUALIFICATION:

No	Minimum Criteria for Qualification	Fulfilled the specified criteria Yes/No
1	**OEM Eligibility (Original equipment manufacturer)**	
	1.1 OEM should be ISO 9001 certified for the Quality Management systems. **Copy of ISO certificate should be attached and ISO certificate should be one year old and validity should be there.**	
	1.2 OEM shall be a Company registered in India under the Indian Companies Act 1956 and existing for the past 5 years as on 31.08.2021. Copy of Registration Certificate as proof of registration should be provided.	
2.	Bidder(Systems Integrator) Eligibility	
2.1.	Establishment of the firm must be 10 years old in the IT Infrastructure business as on 31.08.2021. The **firm** should be registered with Registrar of companies and Sales/Trade tax/Service Tax department. Copies of PAN Number, Sales Tax number, Service tax number along with company registration certificate also with the GST & IGST Registration Certificate should be provided. Profit/loss statement for last three years must be attached. The firm average annual turnover should be more than 6.25 crores for the last 3 years.	
	Firm/Bidder should provide the following details to confirm the above criteria:	
	2.1.1) Provide PAN Number ,GST & IGST details	
	2.1.2) Provide Registration Certificate as proof of registration with Registrar of Companies (Firm must be at least 10 years old as on 31.08.2021)	
	2.1.3) Provide audited Profit/Loss statement for last three years.(Turnover of more than 6.25 crores for the last three years)	
2.2.	The firm/OEM should have good experience in Mass storage (need not be object storage) installation. Firm/OEM should provide 3 PO copies (of which, one PO must be atleast 5 years old, one PO must be of a storage with a capacity of 1Peta Byte and atleast four crore value. PO copies to be attached with installation report and customer details.	
	2.2.1) The bidder/OEM should provide any one of the following with supporting document, ➢ One PO copy of 5 crores with respect to storage installation. ➢ Two PO copies of 3 crores each with respect to storage installation. ➢ Three PO copies of 2 crores each with respect to storage installation.	
3	The bidder/OEM must enclose a customer satisfaction letter from one client preferably from a State/Central Govt.	
4	Authorization Letter from all the OEMs, authorizing the supplier to quote should be attached.	

5	The bidder/OEM should provide at least 10 man power listing to support his technical expertise in storage installation including software, hardware and applications (Engineers details expertise has to be clearly mentioned)	
6	The firm must provide a self-declaration that there is no complaint/vigilance inquiry against them in any Govt./Department /PSU and they have not been black listed by any Govt. Department/PSU.	
7	Provide current financial year solvency certificate worth of 2 crore (Document to be attached).	
8	Agreeing to all Terms and Conditions mentioned in Clause V (Acceptance Criteria) and Clause VI (Terms & Conditions) of the contract	

IV. COMPANY EVALUATION SHEET

The following questionnaire need to be filled and should be signed by a responsible and authorized person of the Company /Agency. Firm should provide only genuine data and any discrepancy found at a later point of time may result in rejection of the bidder from purchase process. Furnishing of data cannot be considered as automatic qualification for participation in the tender.

Sl.No	Description		Vendor Response
1.	Name of the company	:	
2.	Type of the Company (Proprietary/Pvt.Ltd/Public Ltd/Joint Venture/Consortium)	:	
3.	Registration number & certificate	:	
4.	Name & Address of the Office of the Chief Executive of the Company	:	
5.	Contact person for this tender with name & address, contact number and Email address.	:	
6.	Locations of the Branches of Company (if any)	:	
7.	From which year the Company is in operation and year of inception	:	
8.	The Profit & Loss Account details for the last 3 years which is duly audited and submitted as part of the Annual Report. **Kindly fill up the following details without fail.**	:	

Sl.No	Description	Vendor Response		
	in Lakhs only	2017-18	2018-19	2019-20
	Total assets (i)			
	Current assets (ii)			
	Total liabilities (iii)			
	Current liabilities (iv)			
	Net Worth (i-iii)			
	Working capital (ii-iv)			
	Turnover in lakhs			
	Profit/Loss in lakhs			
9.	The major lines of business: :			
10.	Technical Manpower details :			

Description	No. of Persons	Remarks
Engineer		
Supervisor/ Manager		

11.	Major customers :	

12. Details of last 5 years' experience in Supply, Installation, Testing and Comprehensive Warranty support of storage products: Fill up the order details as per attached Purchase Order copies in 2.2 under Bidder eligibility.

Sl.No.	Full postal address of the client	Contact Person name and Email ID	Description of the work	Purchase Order Reference No. with date	Value of the work (Rs. in Lakhs)	Completion Time as per PO
1)						
2)						
3)						

13.	Acceptance to carry out the Supply, Installation, Testing and Comprehensive Warranty support as given in the Tender terms and	**YES/NO**

	conditions	
14.	Acceptance for not sub-contracting the awarded work.	**YES/NO**
15	Acceptance for the terms and conditions listed in our Tender form.	
16	Confirm the submission of PO copy which shows supply and installation is executed for Cluster Systems.	
17	Confirm that the copy of ISO certificate is attached	
18	Confirm that the customer satisfaction letters are enclosed	
19	Confirm that the submission of OEM Authorized Letter from OEM, authorizing the supplier to quote is enclosed.	
20.	Confirm that the Self declaration letter is attached	

Signature of Authorized Person with Seal

V. TERMS & CONDITIONS

1) The scope of the work involves Supply, Installation, Configuration, Testing and Commissioning of Scalable Object-based Storage Solution as per tender specifications in Section VII.
2) One onsite engineer for one year to support after commissioning of Object Storage as per Annexure-II in Section VII and in Price Bid Format.
3) The vendor should clearly mention the make of the quoted items.
4) The vendor should attach the technical brochure for all the items quoted in their offer.
5) The vendor should provide compliance statement for all the specifications of all deliverable items as per Annexure -I. **An offer without compliance statement will not be considered**.
6) **Better Commissioning (Supply, Installation, Configuration, Testing and Commissioning of Scalable Software Defined Distributed Native Object and File Systems Scale-Out Storage Solution) as well as for faster completion of project, single supplier may be selected for all items. Hence, overall lowest may be recommended for all Items.**
7) Installation and commissioning charges (if any) should be clearly mentioned in the tender.
8) The supplier should arrange their own means of transport both for Personnel and installation.
9) The Supplier should observe the security procedures applicable at SDSC SHAR. SDSC SHAR, Sriharikota is declared as Prohibited Place under Official Secrets Act, 1923 and infiltrators into the Premises of SDSC SHAR are liable for prosecution. The supplier is required to ensure that persons of trustworthy bearing good character, discipline and antecedents only shall be deputed to carry out warranty support and take the responsibility for their proper conduct during the period of warranty. The engineers deputed to SDSC SHAR must display his Identity card, issued by the agency along with the entry permit issued by SDSC SHAR while moving in SDSC SHAR.
10) SDSC SHAR will not be responsible for any injury to your personnel, damage to your property, caused at site.
11) The security regulations of the center should be faithfully observed and any loss or damage incurred by us on account of your failure of any nature to follow such security regulations will be to your account.
12) Entry pass will be issued to your service Engineer identified to attend the warranty calls at SDSC SHAR. For this purpose, please furnish the personal details of the service Engineer by email well in advance along with vehicle number. Please note that SDSC SHAR is observing

5 days week i.e. Monday to Friday only and shall remain closed on all Central Govt. and National holidays.

13) SDSC SHAR will provide accommodation for the Engineers / Technicians who are deployed for the Supply, Installation, Configuration, Testing and Commissioning of Scalable Software Defined Distributed Native Object and File Systems Scale-Out Storage Solution, on chargeable basis subject to availability.

14) The vendor should provide training about the configuration and testing procedures followed for the installed Scalable Object-based Storage to the identified SDSC SHAR Engineers / Technicians.

15) Authorization Letter from all the OEMs, authorizing the supplier to quote should be attached

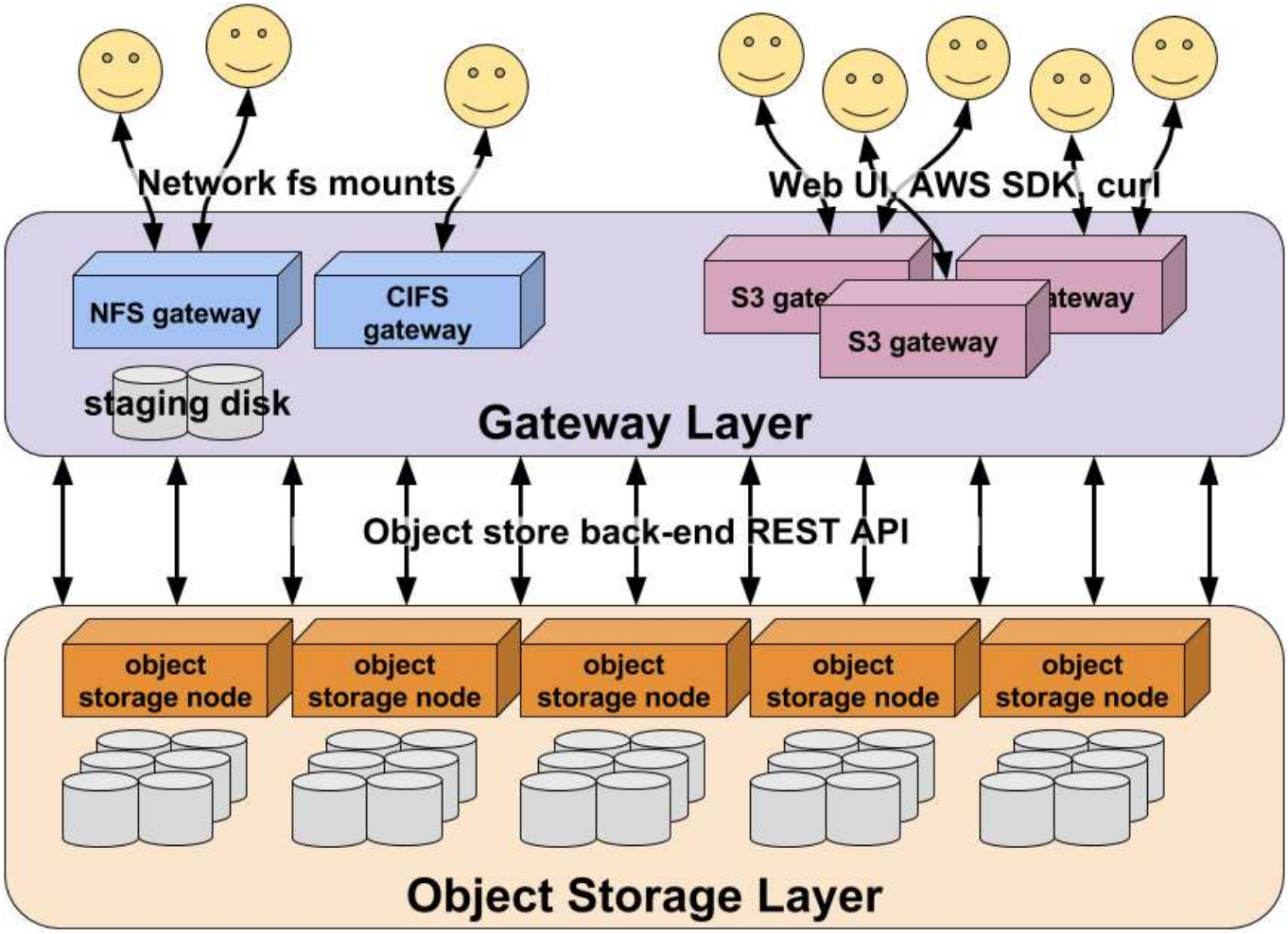
VI. Objective of this RFP:

The objective of this RFP is to deploy and implement Scalable Object-based Storage Solution for Distributed File Systems and Object Storage of 1 Peta Byte (PB) Usable which needs to be stretched across two data centres to provide the below functionalities for users and applications:

- Digital Asset Management
- Storage repository for Surveillance data of recording server for video management software

  (from IP cameras located within designated work areas)
- File Sync and Share (drop box-like) functionality

SDSC SHAR's Requirement for Scalable Software Defined Distributed Native Object and File Systems Scale-Out Storage Solution for 1PB: -

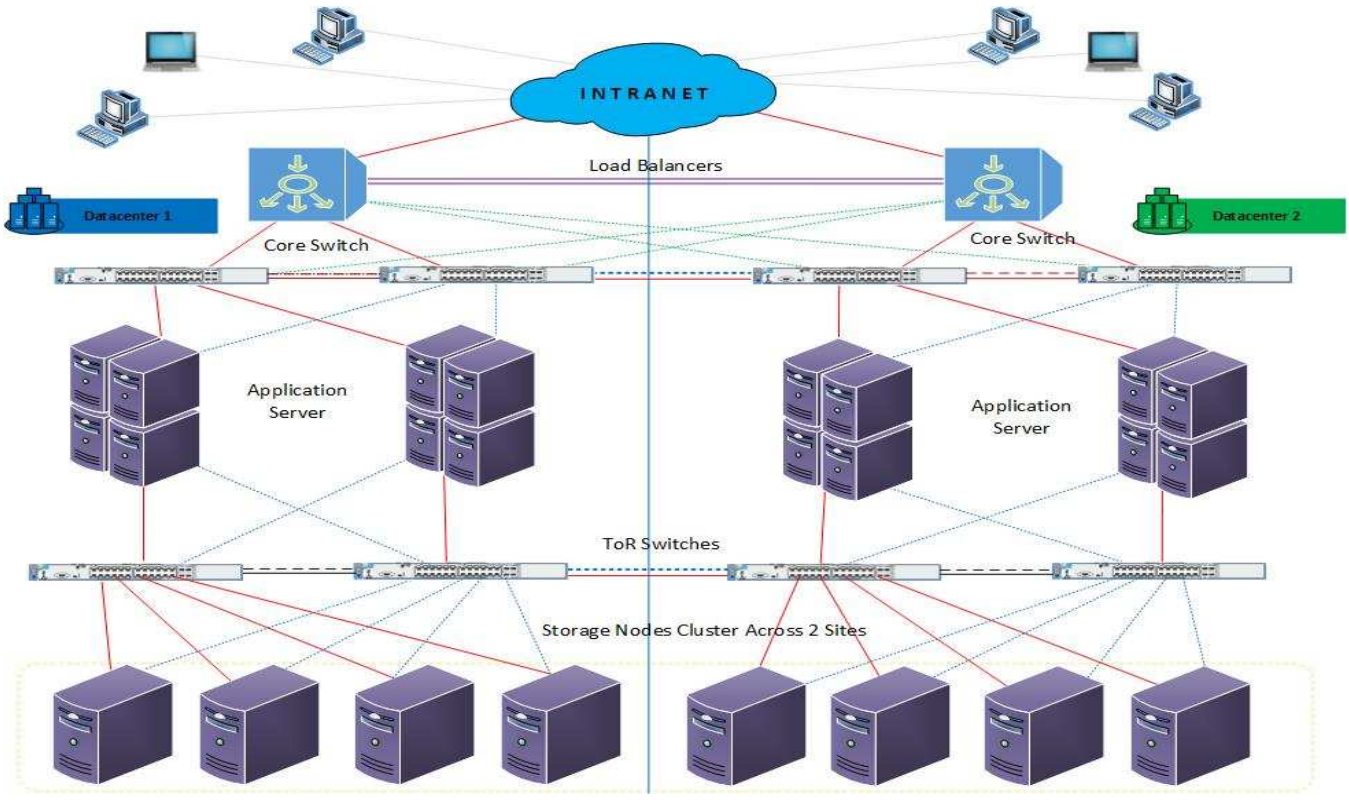


- File Sync and Share (drop box-like) functionality
  - ✓ Primarily for desktops/workstations within Intranet for data protection and unlimited retention
  - ✓ no external access
  - ✓ no access through portable devices (phones, tabs, pen drives etc.)
  - ✓ Currently about 1000 devices/users are to be covered

- ❖ Digital Asset Management
  - ✓ Data immutability with WORM functionality
  - ✓ Primarily for all ‘Unstructured and Semi-structured’ data including documents, spreadsheets, video, etc.
  - ✓ ability to identify and tag data depending on content with the ability to search based on meta-data
  - ✓ File Sizes varying from few-KBs to few-GBs/TBs.
- ❖ Storage repository for Surveillance data (from IP cameras located within designated work areas)
  - ✓ Scaling to about 200 – 250 cameras in HD mode with 15 fps
- ❖ Overall Storage Usable capacity required for all the above-mentioned workload is 1PB useable stretched across multi-site

Solution Overview

Two-Site Stretched View



2-Site Stretched Configuration Assumption: -

- **8 Nodes across 2 Sites**
- **4 Nodes per Site**
- **Each node configured with 312TB capacity (26 x 12TB)**
- **Total Stretched Capacity – 2496 TB**

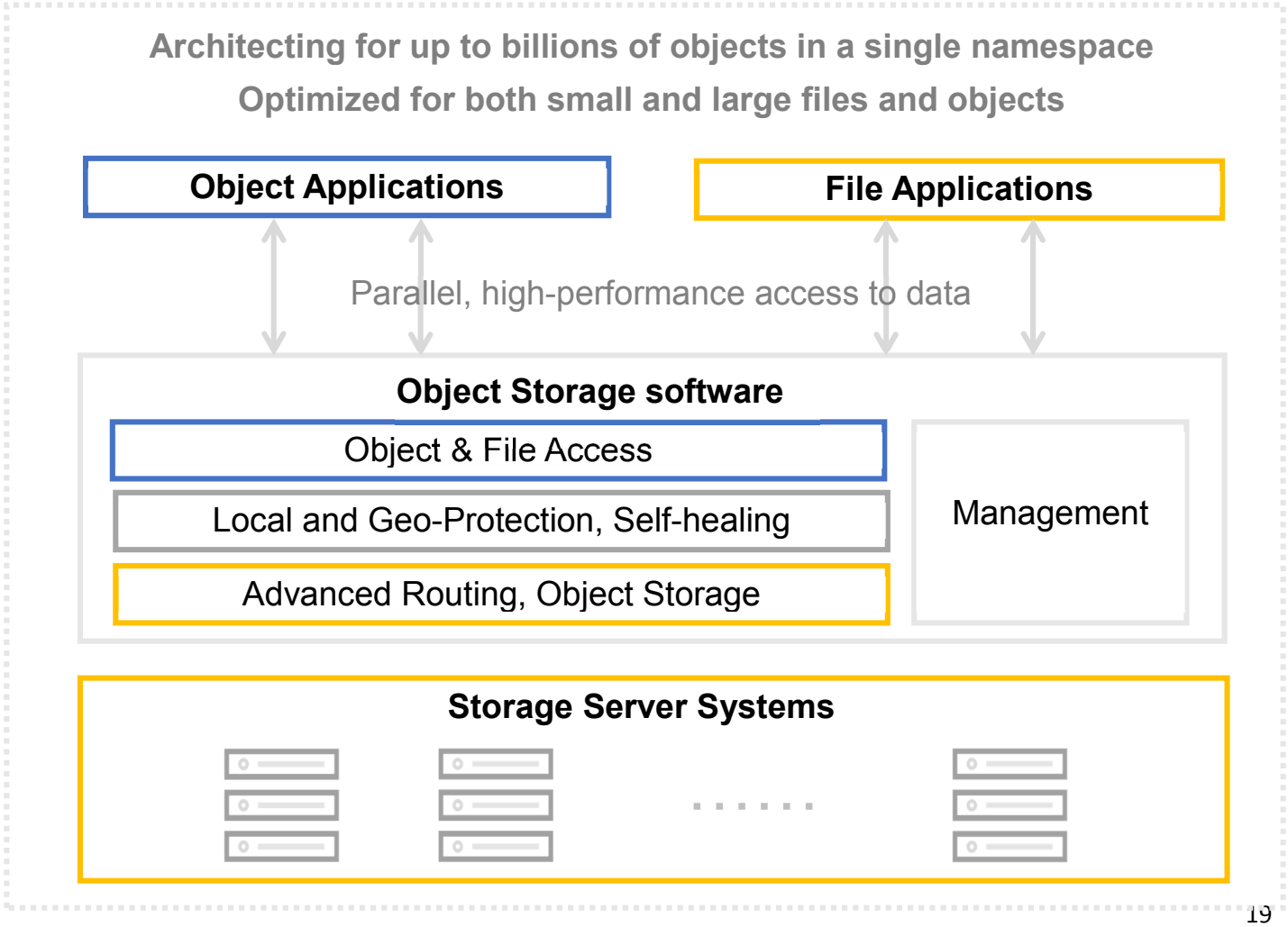
ASSUMPTIONS MADE FOR SIZING	
Weighted average size for files/object I/O (KB)	400
Percent of storage containing EC encoded objects	95%
Site description	Two sites stretched COS (3) EC (5,7)
Number of S3 connectors	5
Number of NFS connectors	0
Number of SMB connectors	3
Storage Server Summary	
Storage Server chassis type	X86 Storage Server
Total storage nodes	8
Drive selection (TB)	12
Drives per storage node	26
Raw storage capacity in TB per storage node	312
Usable storage capacity in TB per storage node	125
Data protection plus file system overhead	148%
Erasure coding scheme	EC (5,7)
Replication scheme	COS 3
Total raw storage (TB)	2496
Total data protection overhead (TB)	1496
Total usable storage (TB)	1000
Total data spindles	208
Number of sites the primary data stored.	2 (two sites stretched) −Supports loss of entire site

*Assumed that 95% of data will be using EC and 5% of data will be using COS.

Note: The number of nodes given above are minimum in each site. More number of nodes in each site with required storage space is acceptable.

Scalable Software Defined Distributed Native Object and File Systems Scale-Out Software

- Completely Hardware Agnostic
- Peer-to-Peer Philosophy
- End-to-End Parallelism
- Internal distributed DB for metadata
- Native Object and file interfaces
- Advanced Data Protection – No RAID
- Simple Operations & Management
- Extended Data Management Features
- Compliance Capabilities – WORM compliance and Various regulation
- Can scale up to 2^160 objects
- Supports mixed workloads with small and large files
- Multi-tenancy and data isolation
- Built-in Load-Balancing for Object (REST, S3) and File (SMB & NFS)
- Self-healing



VII. SPECIFICATIONS

1. Specifications for Scalable Software Defined Distributed Native Object and File Systems Scale-Out Storage Software License for 1Peta Byte, Storage Nodes, Management Nodes, Virtualization software, Enterprise File Synchronization and Sharing Solution Software, Digital Asset Management Software and Server Load Balancer & Global Server Load Balancer as per given in Section VII and Section VIII (Lists of Deliverables).

2. One onsite engineer for one year to support Object storage maintenance as per given in Annexure-II and Bill of Material in Section-X.

Annexure-I

1. Storage Nodes/Server Specification – Minimum of 8 Nodes

Feature	Details
Processors	2 Physical CPUs, minimum Intel Xeon 16 Cores CPU
Memory	192GB ECC DDR4 Memory Kit
Controller	Integrated RAID Controller which supports RAID 0,1,5 & 6 with minimum of 4GB Cache along with Battery Backup Unit Module
Hard drives	2 x 1TB SATA/SAS HDD in RAID 1 Selection of either 10TB/12TB 3.5" 7.2K RPM Helium or better HDDs being used for Data Storage 1.6TB PCIe x8 Lanes Mixed-Use Solid-State Drive Card for storing metadata in high performance
Network	Embedded Dual-Port 1 GbE RJ45 Ports, Dual Port 10 GbE Base-T Ethernet Adaptors x 2 nos (Totally 4 x 10Gbe Base-T Ports)
Expansion	3 slots or more after configuring the above requirements
IO	4 USB
OS	Operating system supporting storage node to be configured and installed.
Powersupply	2 X hot plug power supply (230V @50Hz) unit
Management	System management software to monitor the system health at local console and over network
Accessories	2U chassis with rail kit

Warranty	OEM Vendor Support on 24x7 for 3 Years Onsite support with Defective Media Retention policy to allow the customer to keep their storage medias

Note: The number of nodes given above are minimum in each site. More number of nodes in each site with required storage space is acceptable.

2.Application & Management Servers Specification – Minimum of 4 Nodes

Feature	**Details**
Processors	2 Physical CPUs, minimum Intel Xeon 20 Cores CPU
Memory	512GB ECC DDR4 Memory Kit
Drive Bays	8 SATA / SAS hot pluggable drive bays
Controller	Integrated RAID Controller which supports RAID 0,1,5 & 6 with minimum of 2GB Cache along with Battery Backup Unit Module
Hard drives	2 X 480 GB SSD Drive for RAID 1 configuration
Network	4 X 100/1000 Ethernet (not considering management Ethernet port) and Dual Port 10 GbE Base-T Ethernet Adaptors x 2 nos. (Totally 4 x 10Gbe Base-T Ports)
HBA	Dual port 16 Gbps SAN host bus adapter
Expansion	3 slots or more after configuring the above requirements
IO	4 USB, DVDRW
OS	Virtualization software with production support for 3 years
Power supply	2 X hot plug power supply (230V @50Hz) unit
Management	System management software to monitor the system health at local console and over network
Accessories	2U chassis with rail kit

Warranty	OEM Vendor Support on 24x7 for 3 Years Onsite support with Defective Media Retention policy

3. Secure Enterprise File Synchronization and Sharing Solution Software

Scenario Description:

Enterprise file synchronization and sharing software services to enable users to securely synchronize and share documents like DropBox functionality from their desktop or laptop devices.

Solution Features:

• File Drop • File Locking • File Versioning • File Tagging • Content Sharing • Sharing Permission Management • File Integrity Checking • Large File Support (>4 GB)	• Upload Chunking/Resuming • Active Session Management • Active Directory/LDAP Integration • Intuitive Web-Interface • Should Support Desktop Clients for Windows, MacOS & Linux • Must Support S3 Native Object Storage as a target storage

Self-Hosted	Run EFSS in On-Premises infrastructure, with full control over your data.
Option to Integrate with Existing Fileservers	Integrate with Microsoft Active Directory (AD) or LDAP to synchronize user accounts and groups
NTFS Permissions	Allow remote access to existing Windows File servers preserving user NTFS permissions
Full Text Search	Seamlessly find documents by searching through file content even in your existing file shares
Anti-Virus scanning	Automatic anti-virus scanning of files when uploading. Integrate with any enterprise anti-virus (e.g. Symantec) that supports ICAP
Single Sign-On (SSO)	Integrated SAML 2.0, Microsoft NTLM and Microsoft ADFS login support. SSO for desktop clients
Full encryption of files at rest	Encryption at Rest

Advanced Sharing	Public and Private sharing with expiration dates and other global policies
Large File Support	Upload and share large files with no size limit

4. Digital Asset Management Software

for simple and fast way to organise digital assets

Scenario Description:

Digital asset management systems needed for creating Library or archive for bulk storage of infrequently changing video or photo assets with the key management operations functionalities on digital assets like Creation, Indexing, Workflow, Version control and Access control and options for in-app customization.

Supporting different resource types of files getting uploaded by the designated users. And allowing the users to 'tag' all files they upload using the template set up that can be descriptive metadata tags like 'colour' and 'product' or workflow tags like 'date' and 'owner'. Users will search using the metadata descriptions - making sure results are consistent and reliable.

Allowing to Easily embed contents/assets inside documents and sites. Also, making collaboration happen by easily organising, sharing and distributing assets among users and departments.

Solution Features:

• Upload & Store Features • File Preview • Easily search • Asset Metadata • Version Control • Upload assets singularly or in bulk • Share & Distribute Features • Grouping assets • Embedding Your Media • Tenants, Groups & Users wise access	• Watermarking • Sharing Assets Securely • Insightful Reporting • Automatically Tag Assets • Intuitive, modern web-based user interface • Workflow Automation • Security • Enterprise Single Sign-On • LDAP directory integration • RESTful API Integration

5. Server Load Balancer & Global Server Load Balancer on HA

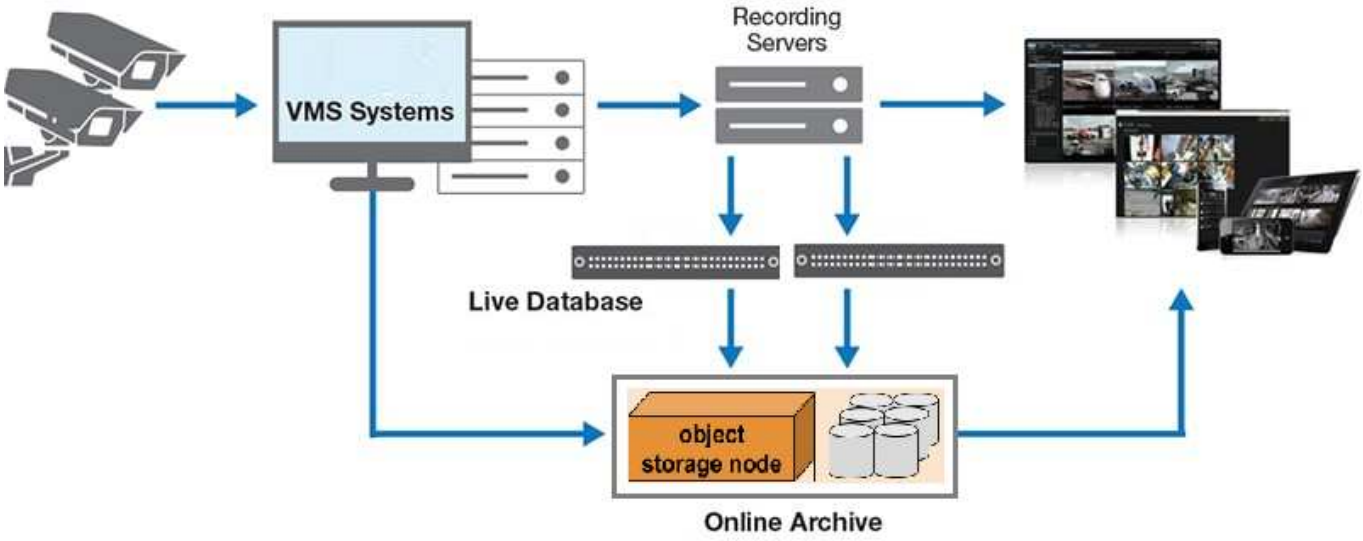
Features	Description
Availability Features	Layer 2-3 Load Balancing, Layer 2-7 Policy & Group Management, Layer 4 Load Balancing, Server Persistence, Global Server Load Balancing, Link Load Balancing, Route L4 Routing, Application, Server & Link Health Checks, Clustering up to 32 nodes, policy L7 Application Scripting
Networking Features	Dual Stack Lite – IPv6 to IPv4 and IPv4 to IPv6 NAT and full IPv6 addressing, Link aggregation, VLAN/MNET, NTP – Static and port-based NAT, advanced NAT for transparent use of multiple WAN links
Acceleration features	Application Performance - Dynamic detect – Client connection persistence – Connection multiplexing – TCP buffering, SSL Acceleration (2048 & 4096-bit), Caching, Traffic Shaping
Security	WebWall Web Application Security - Hardened OS – Secure access only, access control based on client certificate information and access method,DDoS Protection (SLB),SSL Intercept - L2 or L3 mode, integrated or distributed mode, forward or reverse proxy mode – Web agent service, Client-Server Certificate Management, Client Certificate Authentication & Authorization, Client Certificate Application Integration, Secure Application Access - AAA support for Security Assertion Markup Language (SAML), LDAP, RADIUS and Open Authorization (OAuth) protocols
Management	AAA support for Security Assertion Markup Language (SAML), LDAP, RADIUS and Open Authorization (OAuth) protocols

6. Scalable Archive Storage Solution for the Video Surveillance Data

Scenario Description:

- Providing a scalable software-defined native object-based storage solution that will run on industry standard server platforms, which will be match the scalability of the VMS software running.
- The solution will be providing as a secondary tier of storage that is redundant by architecture to the VMS Recording Servers.
- The solution will be highly scalable since both the VMS and Native Object-based SDS solutions can expanded limitlessly on a server-by-server basis infrastructure.
- Long retention periods mandated by compliance regulations

Solution Architecture:



Annexure-II

Onsite Support Services for a period of 1 year.

1. One Onsite first class Graduate engineer in ECE/ Computer science/ E&I with 3 years' experience, and having 2 years experience in handling storage systems should be posted at SDSC SHAR continuously for a period of one year after completion of Installation and acceptance of the system.

2. Onsite engineer should be capable of handling object based storage system independently.

3. **Scope of Work for onsite Engineer:**
   - Monitor & manage the Object Storage System and other software mentioned in the PO
   - Diagnose and fix the problem in cluster as reported by users.
   - Manage vendors and raise incidents with respective vendors.

4. Generally onsite Engineer has to give technical support for 5 days (Monday to Friday). In case of any urgent need during out of office hours/ Launch activities onsite engineer has to give support without any additional cost.

5. Onsite engineer may be provided shared accommodation at Sullurpeta subject to the availability.

6. Onsite engineer has to make his own arrangement for commuting to SDSC SHAR, Sriharikota.

7. Onsite engineer should have a two wheeler with valid license to commute inside the SDSC SHAR.

8. In case of non-availability of concerned engineer during warranty period an amount quoted for man power deployment will be deducted accordingly on pro rata basis.

VIII. List of Deliverables:

1. <u>Scalable Software Defined Distributed Native Object Storage and File Systems Scale-Out Software or Solution: -</u>

Sl. No.	Specifications	Fulfilled the Specified Criteria YES / NO
	Software Defined Storage Features	
1	**Completely Hardware Agnostic**	
	• Uses standard x86 servers, even mixed configurations	
	• Flexible OS support with Linux distribution: CentOS 7.x and RedHat 7.x	
	• Heterogeneous multi-model/multi-manufacturer infrastructure possible within the same namespace	
	• Ability to mix different size drives (8TB or 10TB or 12TB or Higher capacity Helium or better Hard Drives only) in the same cluster	
2	**Object Storage as a Core Technology**	
	• Object storage technology abstracts underlying storage servers, to create a uniformly scalable storage pool	
	• No size limit for object or files (sparse file technology)	
	• Solution that can scale up to 2^160 objects	
	• Automatic Rebalancing of existing and new data when adding a new storage nodes/server in the cluster	
	• Capacity expansion scaling to be delivered with scale-out / Scale-up architecture	
3	**Peer-to-Peer Philosophy**	
	• Topology with P2P Erasure Coding architecture and inter-routing mechanism	
4	**End-to-End Parallelism**	
	• High level of concurrency from clients to storage nodes	
	• Petabyte-scale: Scale out capacity and performance independently without limiting no. of objects stored	
5	**Internal distributed DB for metadata**	
	• No Metadata controller needed	
	• No Single point of failure	
6	**RICH CONNECTIVITY**	
6.1	**Native Object interfaces**	
	• Object interfaces: HTTP/REST, S3 highly compatible API, SWIFT	
	• Scalable S3 Erasure Coding architecture including the following:	
	o S3-Server: S3 API Server for Buckets/Objects, MPU and more	
	o Scale-Out "any-to-any" access	
	o **Security model** S3-Vault: Security service for Accounts	
	- Multi-tenant, Support for S3 IAM – Identity and Access Management	
	- Microsoft Active Directory over SAML 2.0 (ADFS) Integration	
	- Comprehensive IAM security model for Users & Groups with Roles	

	- Bucket & Object ACLs	
	o S3-Metadata: Distributed Metadata Engine	
	o S3 Bucket Service Utilization API (UTAPI) + Account level Utilization metrics	
	o S3 Bucket Versioning API	
	o Transparent Bucket-Level At-REST Encryption	
	o S3 Stretched deployments for 2 & 3-sites	
	o S3 Cross Region Replication for asynchronous replication	
	o S3 Console: GUI Web interface to manage accounts, users, policy and monitor usage	
	o S3 Browser: GUI Web interface to create buckets and upload objects	
	o Quota for S3	
6.2	• **File interfaces:** SMB(CIFS), NFSv3 and v4	
	o Capability to write simultaneously to the same folder from different file connectors (folder scale out)	
	o Seamless load balancing and failover among the file connectors	
	o Federated Access “Single Sign On” to S3 Connector	
	o Compatible with LDAP, Active Directory and Kerberos	
	o Volume protection feature (can’t modify/delete files once written to the volume)	
	o Quota for file	
	o Supports mixed environment Windows & Linux	
	• **Bridge object/file**	
	o Global and unlimited namespace and concurrent data access from various clients	
	o **All File interfaces (NFS, SMB, Linux FS and CDMI) share a common namespace: write with one, read with the other**	
	o NFS to S3 compatibility (write in NFS read in S3 and write in S3 read in NFS)	
6.3	• **Multi-tenancy and data isolation**	
6.4	• **User, Account and Bucket metrics data collection via API and GUI Dashboard**	
6.5	• **Mixed workload with Small and Large files support**	
6.6	• **The offered solution should include load-balancing for Object (REST, S3) and File (SMB & NFS) with necessary redundancy**	
7	**RELIABILITY**	
7.1	**Advanced Data Protection – No RAID**	
	• The multi-site stretched cluster solution should provide a minimum of 11* Nines of availability	
	• Providing data protection and resiliency through 2-Site /3-Site Geo-Stretched/Geo-distributed erasure-coding and replication	
	• Highly-available disaster-recovery enabled deployments through multi-data center deployments to provide availability in the event of full site-level failure	
	• Erasure Coding for large files (>60 KB or adjustable)	
	• Replication ratio up to 6 copies for small files (<60KB or adjustable) / Erasure coding	
	• Possibility to adjust the Replication ratio & the Erasure Coding ratio depending on the file size	

	• Very flexible Erasure Coding ratio (5,7) or (7,5)	
	o Flexible data protection mechanism can be configured for each application or usage (classes of services)	
	• Rack-aware - Full independent object placement across racks	
7.2	• **Flexible topology**	
	o Min. support of Multi-Geo up to 3 sites (object level)	
7.3	• **Self-healing**	
	o File integrity checking when reading the file and automatic rebuilt if an error is detected (e.g. bit rot)	
	o Each object is linked with a minimum of 3 components protection. A loss of 3 hard drives doesn't cause data loss	
8	**EASE OF USE**	
	Full management and control with Supervisor and CLI	
	o Full Topology View: Storage Cluster status with Zone, Server & Node details	
	o Graphing capabilities & dashboards	
	o SNMP MiB Browsing, Traps and Graphical Usage Monitoring	
	o REST APIs for monitoring & management	
	o Role Based Access Control (RBAC)	
9	**Simple Operations & Management**	
	o Software Upgrades, Server replacements, or Capacity Extensions don't stop the system	
	o Automated disk failure detection	
	o Automatically rebuild for failed drive	
	o Automated storage rebalancing in the event if capacity or nodes upgrade	
	o The system has to automatically rebuild the missing data in case of a hardware component failure	
10	**EASE OF INSTALLATION**	
	• Automation option available for installation	
	• Volume Provisioning User Interface: Should be able to provision & mount the storage as a volume	
11	**Extended Data Management Features**	
	• S3 metadata and tag search	
	• Lifecycle: automatic transition and expiration of data based on criteria	
	• Replication one to many (asynchronous bucket replication to several s3 targets)	
12	**COMPLIANCE CAPABILITIES**	
12.1	• Requirements for a compliant solution are based on various regulations e.g. SEC, GDPdU, GoBS, Euro-SOX, AO, Basel III, SOX, HIPAA ...	
12.2	• Data must be tamper-proof	
12.3	• Data cannot be deleted (**WORM** like)	
12.4	• Data must be kept for a specified period (retention)	
12.5	• Data must be able to be migrated to an alternative media	
12.6	• Access to individual archived objects must be possible at all times	
12.7	• Data replication and integrity protection	
13	**SUPPORT**	
	• 3 years 24x7 OEM/Vendor Support	

2. Storage Nodes/Servers Specifications for Native Object Based Storage configuration should have minimum of:– 8 Nos

Sl. No.	Specifications	Fulfilled the specified criteria YES / NO
1.	**Storage Servers Specifications for Native Object Based Storage – Minimum of 8 Nos.**	
	• 2 Physical CPUs, minimum 16 Cores of Intel Xeon Processor each	
	• Minimum of 192 GB DDR4-2666 Registered Smart Memory Modules	
	• The OS running the Object Storage solution must be separated in RAID 1 mirrored HDDs with minimum of 1TB Drives	
	• 10TB / 12TB Capacity 3.5" 7.2K RPM HELIUM HDDs being used for Data Storage	
	• Integrated RAID Controller which supports RAID 0,1,5&6 with minimum of 4GB Cache along with Battery Backup Unit module	
	• Bidder should provide OS for the Storage Nodes.	
	• Minimum 1.6TB PCIe x8 Lanes Mixed-Use Solid-State Drive Card for storing metadata in high performance	
	• 2 x Dual Port 10GbE T-Based PCIe Ethernet Adapter Cards (Totally 4 x 10Gbe Base-T Ports)	
	• 2 x 100/1000 Ethernet (not considering management Ethernet port)	
	• 2 x hot plug power supply (230V @50Hz) unit	
	• 2U Rack Server with Sliding Rail Kit for mounting the Server Hardware	
	• System management software to monitor the system health at local console and over the network	
	• OEM Vendor Support on 24x7 for 3 Years Onsite support with Defective Media Retention policy	

3. Servers to run Virtualisation software with HA in each site for provisioning VMs to API Connectors, Applications and Management Nodes Specifications:– 4 Nos

Sl. No.	Specifications	Fulfilled the specified criteria YES / NO
1.	**Servers to run Virtualisation software with HA in each site for provisioning VMs to API Connectors, Applications and Management Nodes Specifications etc. – Minimum of 4 Nos.**	
	• 2 Physical CPUs, minimum 20 Cores of Intel Xeon Processor each	
	• Minimum of 512 GB DDR4-2666 Registered Smart Memory Modules	
	• The OS running the solution must be separated in RAID 1 mirrored HDDs with minimum of 480 GB SSD Drives	
	• Integrated RAID Controller which supports RAID 0,1,5&6 with minimum of 2GB Cache along with Battery Backup Unit module	

	• Minimum of 3 PCIe slots MUST be free after populating all the add-on cards	
	• 4 x 100/1000 Ethernet (not considering management Ethernet port)	
	• 2 x Dual Port 10GbE T-Based PCIe Ethernet Adapter Cards (Totally 4 x 10Gbe Base-T Ports)	
	• 1 x Dual Port 16Gbps FC Host Bus Adaptor with Transceiver modules	
	• 2 x hot plug power supply (230V @50Hz) unit	
	• 2U Rack Server with Sliding Rail Kit for mounting the Server Hardware	
	• System management software to monitor the system health at local console and over the network	
	• OEM Vendor Support on 24x7 for 3 Years Onsite support with Defective Media Retention policy	

4. <u>Virtualization Software for Application & Management Nodes should be provided with – 8 Nos</u>

Sl. No.	Specifications	Fulfilled the Specified Criteria YES / NO
1	Virtualisation software with three years Production support.	

5. <u>Enterprise File Synchronization and Sharing Specifications: -</u>

Sl. No.	Specifications	Fulfilled the specified criteria YES / NO
1	Enterprise File Synchronization and Sharing Software Solution in 2-site implementation	
2	File Drop - Allows users to mark any folder as a drop folder for externals to anonymously drop files in a shared folder without revealing the actual content of the folder.	
3	File Locking - Lock files to prevent users from making concurrent edits that could cause file conflicts. Users are notified when files are locked	
4	File Versioning - Versioning-Feature for files which allows rollback to older file versions	
5	File Tagging - Add specific tags to your files. This feature allows smart filtering of files and works with shared files as well.	
6	Content Sharing - Sharing of files and more with users via an anonymous link (password protection, expiration date possible)	
7	Sharing Permission Management - Feature to selectively restrict access to shared content (Create, Edit, Delete, Reshare)	
8	File Integrity Checking - To prevent file corruption the integrity of up- and downloaded files is automatically verified by comparing their unique checksums before and after transfer	
9	Large File Support (>4 GB)	
10	Upload Chunking/Resuming - Chunking allows to split up files into smaller packages (‘chunks’) before uploading. This enables parallel uploading for a	

	single file (performance improvements) and upload resuming during e.g. unstable network connection	
11	Active Session Management - List active sessions of a user and disconnect them selectively.	
12	Active Directory/LDAP Integration	
13	Intuitive Web-Interface	
14	Should Support Desktop Clients for Windows, MacOS & Linux	
15	Must Support S3 Native Object Storage as a target Storage	
16	Support for 3 Years	

6. Digital Asset Management Software Specification: -

Sl. No.	Specifications	Fulfilled the specified criteria YES / NO
1	Digital Asset Management Software in 2-site implementation	
	Support for 3 Years	
	Must support File Formats - images (in file formats including: bmp,eps, gif, jpeg, raw, tiff), documents (in file formats including: xlx, xlxs, ppt, pptx, doc, docx, pdf), videos (in file formats including: Divx, DV, FLV, HDV, h.264, MPEG-2, MPEG-4, SWF, QuickTime, Windows Video, XDCAM) and audio (in file formats including: wav, wma,mp3).	
2	**Upload & Store Features**	
2.1	File Preview - Most file types have to be viewed without having to download them first	
2.2	Need to Easily search across all assets & even inside PDFs	
2.3	Asset Metadata - Show all associated metadata right next to the digital asset	
2.4	Version Control - Full version control of every asset, with data logging & reporting	
2.5	Upload assets singularly or in bulk	
3	**Share & Distribute Features-**	
3.1	Grouping assets easily into multiple collections, no matter the filetype	
3.2	Embed Your Media - Embed your assets across the web with embed code straight from the DAM	
3.3	Tenants, Groups & Users - Only giving specific users or organisations access to certain assets & time limit the access	
3.4	Watermarks - Setting assets to be automatically watermarked to add extra level of security	
3.5	Sharing Assets Securely - Sharing assets securely with no file size limitations & revocable access	
4	**Reporting & Management Features**	
4.1	Insightful Reporting – Should be able to see how often individual assets are being downloaded	

4.2	Continuous Metadata Improvement – Should be able to run a report showing all of the search phrases used to find assets, and searches	
4.3	Reporting on all assets that haven't been viewed or downloaded	
4.4	Automatically Tag Assets - images are automatically analysed and relevant metadata suggested	
4.5	Intuitive, modern web-based user interface	
5	**Workflow automation**	
5.1	Basic workflow process in place involving user submission, approval, through to live then archived	
5.2	Workflow states-Monitor the status of any given resource such as live or archived and determine who can see and edit the content	
6	**Security**	
6.1	Enterprise Single Sign-On using Simple SAML plugin	
6.2	LDAP directory integration, RESTful API Integration	
6.4	Configure for multiple user privacy	
6.5	Monitor and oversee the viewing and use of your assets. Assess and approve user uploads, allocate access, and generate secure URLs for third party sharing	

7. <u>Server Load Balancer & Global Server Load Balancer Specifications: – 2 Nos</u>

Sl. No.	Specifications	Fulfilled the specified criteria YES / NO
1	Server Load Balancer & Global Server Load Balancer Hardware Appliance – 2 Nos.	
2	Should be Hardware appliance-based solution with 64-bit Speed Core Architecture & purpose-built hardware for high performance.	
3	The Appliance should support 4x1GbE copper ports, 2x10GbE SFP+ ports & dual power supply	
4	Should support Minimum 20Gbps Throughput and 10 Gbps SSL throughput with dedicated SSL card to support Minimum 14,000 ECC TPS (ECDSA-SHA256) & 20,000 RSA SSL TPS 2K Keys	
5	The solution should able to load balancer both TCP and UDP based applications with layer 2 to layer 7 load balancing including WebSocket and WebSocket Secure.	
6	The solution should support server load balancing algorithms i.e. round robin, weighted round robin, least connection, Persistent IP, Hash IP, Hash Cookie, consistent hash IP, shortest response, proximity, SNMP, SIP session ID, hash header etc.	
7	The solution should support Multi-level virtual service policy routing – Static, default and backup policies for intelligent traffic distribution to backend servers	
8	The solution using e-policy then should support algorithms including round robin, least connections, shortest response, persistence ip, hash ip, hash ip and port, consistent hash ip and SNMP	

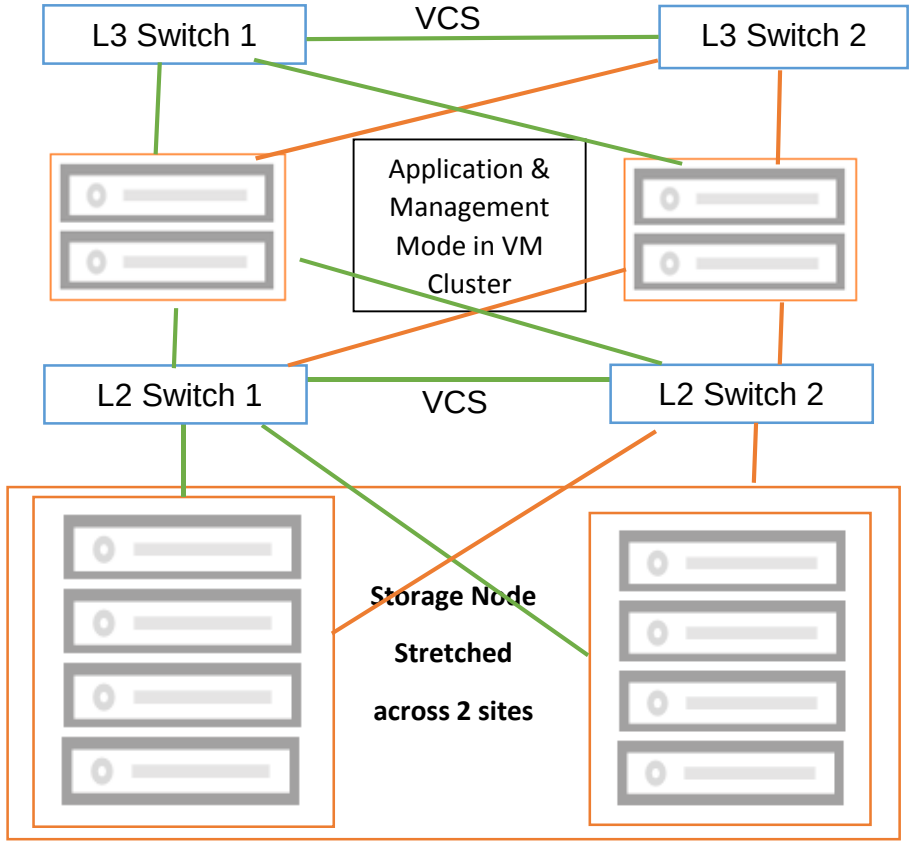
9	The solution should provide application & server health checks for well-known protocols such as ARP, ICMP, TCP, DNS, RADIUS, HTTP/HTTPS, RTSP etc..	
10	The solution should provide compressive support for IPv6 functions to help with ipv4-to-ipv6 transition without business disruption and must provide support for dual stack, DNS64, NAT 64, DNS 46, NAT 46, IPv6 NAT	
11	The solution should support site selection feature to provide global load balancing features for disaster recovery and site redundancy.	
12	It should support advance functions Authoritative name sever, DNS proxy/DNS NAT, full DNS server with DNSEC, DNS DDOS, application load balancing from day one. It should be capable of handling complete Full DNS bind records including A, MX, AAAA, CNAME, PTR, SOA etc.	
13	The solution should able to evaluate round trip time (RTT), Persistence loss ratio (PLR) and hop count for dynamic proximity calculations. It should support dynamic proximity and static proximity rules to direct the traffic to closest datacentre	
14	The solution should support global server load balancing algorithms including - Weighted round robin, Weighted Least Connections, Administrative Priority, Geography, Proximity, Global Connection Overflow (GCO), Global Least Connection (GLC), IP Overflow (IPO)	
15	The solution should provide full IPv6 functionality with support for AAAA DNS resolution, proximity rule support for IPv6, dynamic proximity (to detect IPv6 local DNS's) and SDNS health check for IPv6 service IP's	
16	The solution should support extensible policies (epolicies) TCL scripts to implement business logic on network without changes in application code.	
17	The solution should provide comprehensive and reliable support for high availability and N+1 clustering based on Per VIP based Active-active & active standby unit redundancy mode.	
18	the solution should support Stateful session failover with N+1 clustering support when deployed in HA mode. The solution should support USB based FFO cable to synchronize configuration at boot time of HA	
19	The appliance should have extensive reporting and logging with inbuilt tcpdump like tool and log collection functionality. The appliance should have SSH CLI, Direct Console, SNMP, Single Console per Cluster with inbuilt reporting.	
20	The appliance must support multiple configuration files with 2 bootable partitions for better availability and easy upgrade / fallback. The system should support led warning and system log alert for failure of any of the power and CPU issues	
21	Should support industry standard VRRP for cluster failover	
22	3 Years 24x7 OEM Support with Advanced On-Site Replacement Warranty	

General

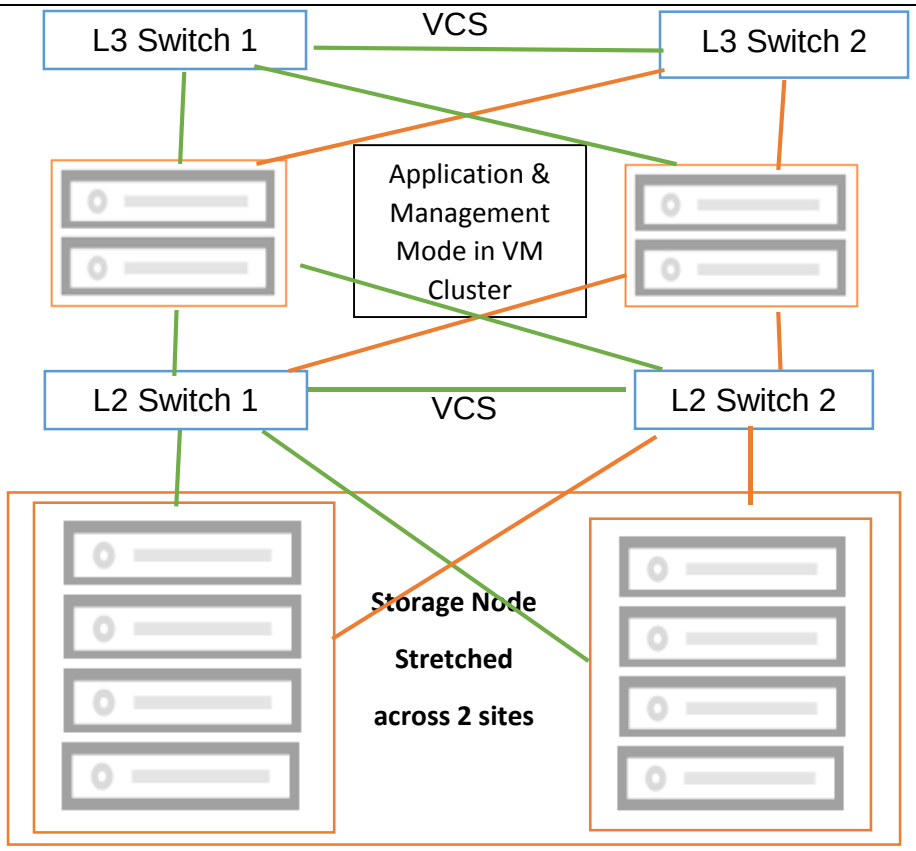
- **The vendor should clearly give make and model number for each item.**
- **Give compliance statement for the above specifications**
- **Vendor should attach the technical literature of the proposed model.**
- **An offer without compliance statement will not be considered.**
- **In case of any failure of hard disk in any of the system during warranty period, vendor has to replace the hard disk. However, the failed hard disk will not be returned to the vendor as per the IT security policy of ISRO**

IX. Site Acceptance Test Procedure:
The L3 Switch-1 and Switch-2 & L2 Switch-1 and 2 will be configured using VCS configuration to behave as a single switch.

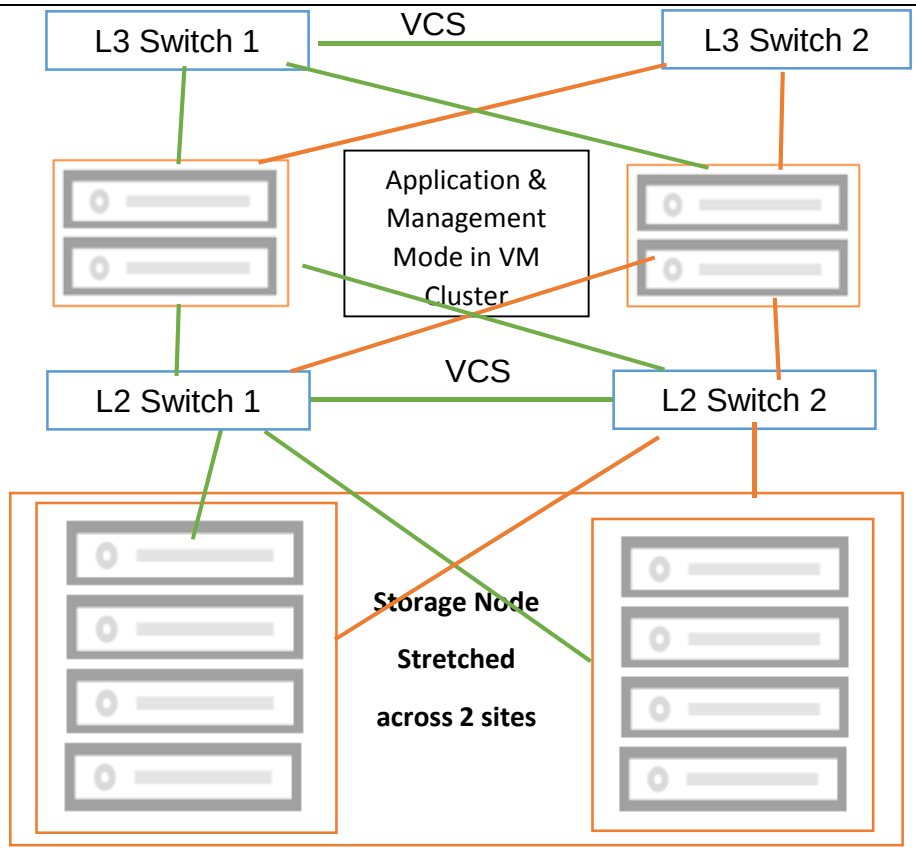
1. Storage Node Stretched Cluster across the site with single site failure: -

Test Item	Storage Node Stretched Cluster across the site with single site failure
Test purpose	2 site Stretched cluster reliability and redundancy during catastrophic failure of one of the sites.
Test Environment	
Test procedure	➢ Test scenario should be performed with the recommended configuration with the test date. ➢ Shut down one of the sites.
Acceptance Criteria	Should Tolerate the loss of single site without Users & Application data access Interruption
Test Results	**Pass** ☐ **Fail** ☐
Remarks	

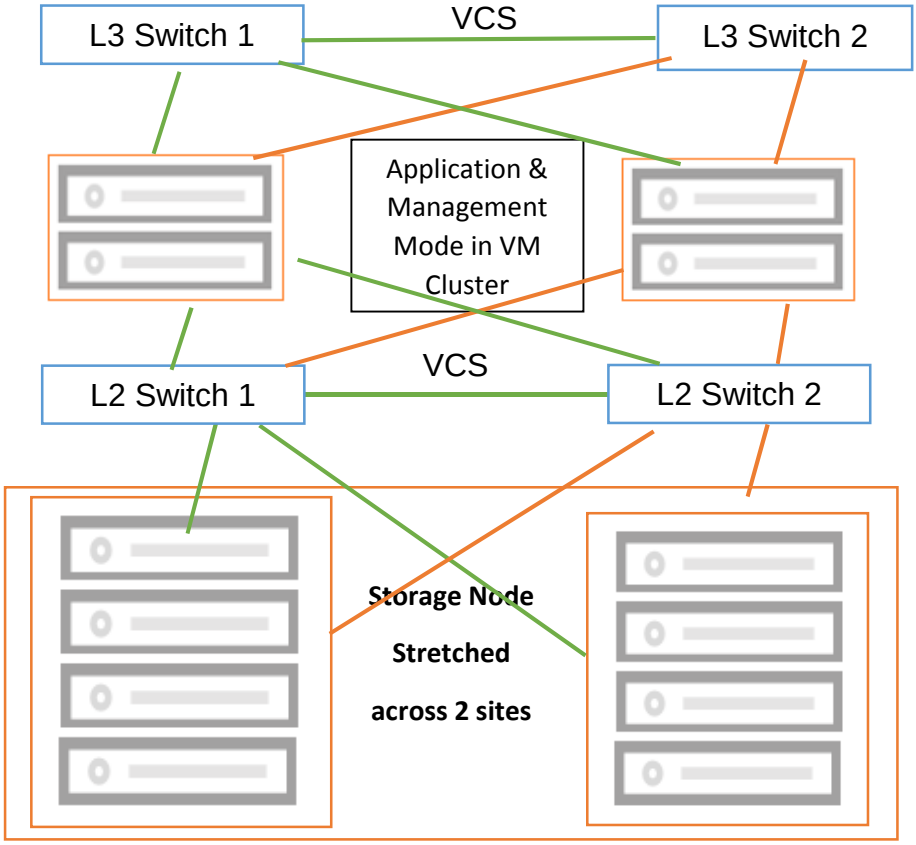
2. Data rebalancing after the site becomes active

Test Item	Data rebalancing after the site becomes active
Test purpose	Rebalancing of data across the storage nodes and its disk drives within the cluster.
Test Environment	
Test procedure	➢ Power ON the site/devices in site-I, which were shut down as a part of Test Case-1.
Acceptance Criteria	Monitoring the data rebalancing and then validating the data availability & accessible to the User & Application after Site becomes active
Test Results	**Pass** ☐ **Fail** ☐
Remarks	

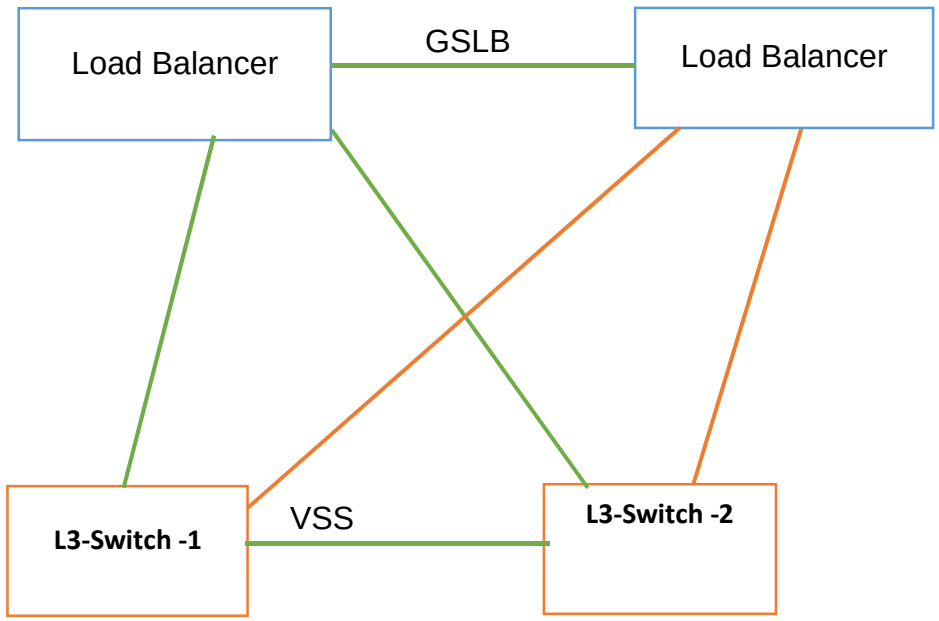
3. Storage Node Stretched Cluster across the site with single Storage/ Storage node failure: -

Test Item	Storage Node Stretched Cluster across the site with single Storage/ Storage node failure
Test purpose	2 site Stretched cluster reliability and redundancy during catastrophic failure of one of the Storage/Storage nodes
Test Environment	
Test procedure	➢ Test scenario should be performed with the recommended configuration with the test date ➢ Shut down one of the Storage/Storage nodes in any of the site.
Acceptance Criteria	Data must be accessible to application gateway & Users without making site failure
Test Results	**Pass** ☐ **Fail** ☐
Remarks	

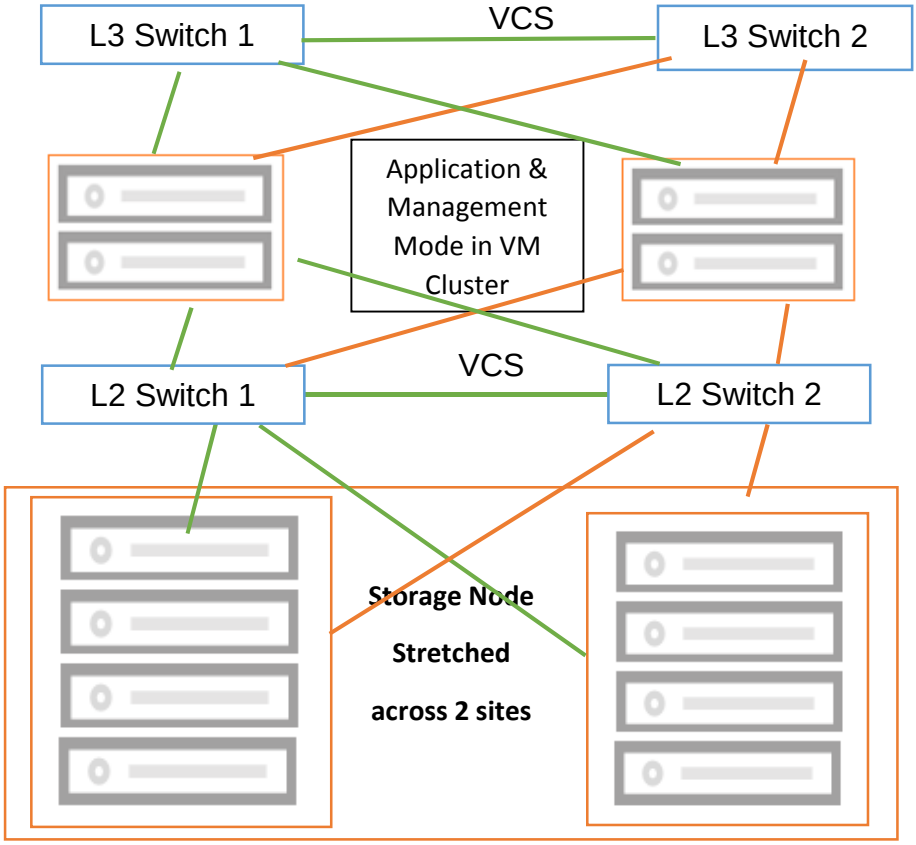
4. WORM Functionality Test: -

Test Item	WORM Functionality Test
Test purpose	To meet the compliance regulation Purpose
Test Environment	
Test procedure	➢ Create a Test volume & assign the WORM information lifecycle management policy. ➢ Populate the Test data with multiple days & weeks WORM information lifecycle management policy
Acceptance Criteria	WORM functionality accessed Volumes data should not be able to Modify or Delete by any level of the privileged, superuser, root or system user and Domain administrator user.
Test Results	**Pass** ☐ **Fail** ☐
Remarks	

5. Load Balancer, Global Server Load Balancer Cluster Failover Test: -

Test Item	Global Server Load Balancer for active-active clustering failover test.
Test purpose	Stack solution reliability and redundancy during catastrophic failure of one of the Load Balancer.
Test Environment	
Test procedure	➢ Initiate flow ICMP to the targeted clustered Load Balancer ➢ Shut down one of the Load Balancer.
Acceptance Criteria	ICMP traffic with minimal disruption of less than 100ms.
Test Results	**Pass** ☐ **Fail** ☐
Remarks	

6. DAM Functionality Test:

Test Item	Digital Asset management software Functionality Test
Test purpose	To meet the compliance regulation Purpose
Test Environment	
Test procedure	➢ Create a User with the Quota. ➢ Create Folder with Files and permissions.
Acceptance Criteria	DAM Features like Sharing and permission to the shared users.
Test Results	**Pass** ☐ **Fail** ☐
Remarks	

SITE TESTS ACCEPTANCE SUMMARY

S. No	Site Acceptance Test Scenarios	Result		Remarks
		Pass	Fail	
I	Storage Node Stretched Cluster across the site with single site failure			
II	Data rebalancing after the site becomes active			
III	Storage Node Stretched Cluster across the site with single Storage/ Storage node failure			
IV	WORM Functionality Test			
V	Global Server Load Balancer Cluster Failover Test			
VI	DAM functionality			

X. PRICE BID FORMAT:

Bill of Materials:

Sl. No.	Item Description	Qty	Basic cost (in Rupees)	Tax (in Rupees)	Total (in Rupees)
1	Supply and installation of Scalable Software Defined Distributed Native Object and File Systems Scale-Out Storage Software License for 1PB usable capacity as per RFP Document.	1 Set			
2	Supply and installation of Storage Nodes/Servers as per RFP Document.	8 Nos			
3	Supply and installation of Servers to run API Connectors, Applications and Management Nodes/Server as per RFP Document.	4 Nos			
4	Supply and installation of latest virtualisation software with three years Production support as per RFP Document.	8 Nos			
5	Supply and installation of Secure Enterprise File Synchronization and Sharing Solution Software for 1000 Users as per RFP Document.	1 Set			
6	Supply and installation of Digital Asset Management Software Specification as per RFP Document.	1 Set			
7	Supply and installation of Server Load Balancer & Global Server Load Balancer as per RFP Document.	2 Nos			
8	Installation, Testing and commissioning charges for all the above items	1 Set			
9	Onsite Support Services for one year.	1 Set			
Grand Total:					

The break up details of the above items are to be given only in price bid and should not be attached in technical bid attachment. Same should be uploaded as an attachment in price bid.